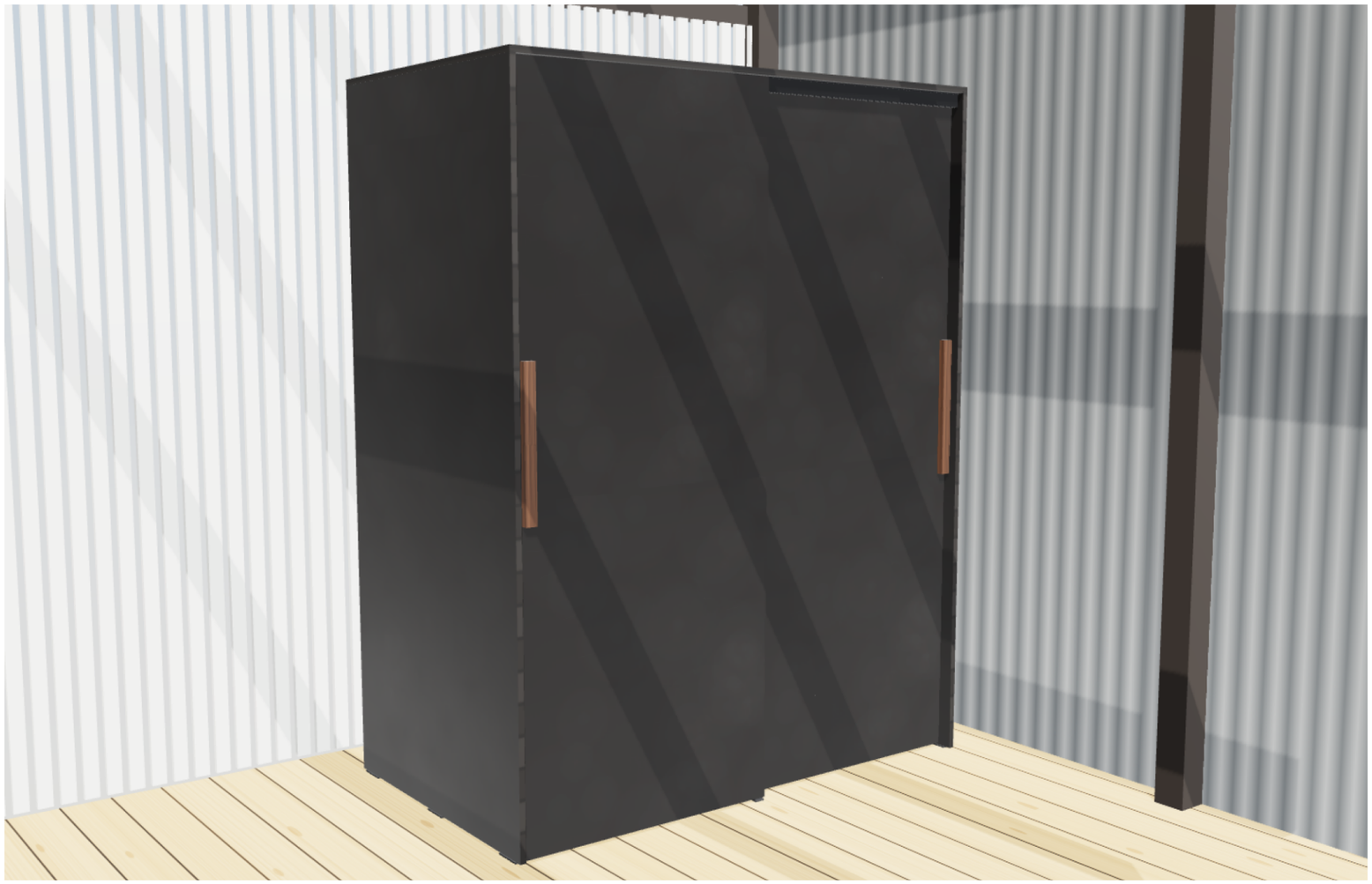
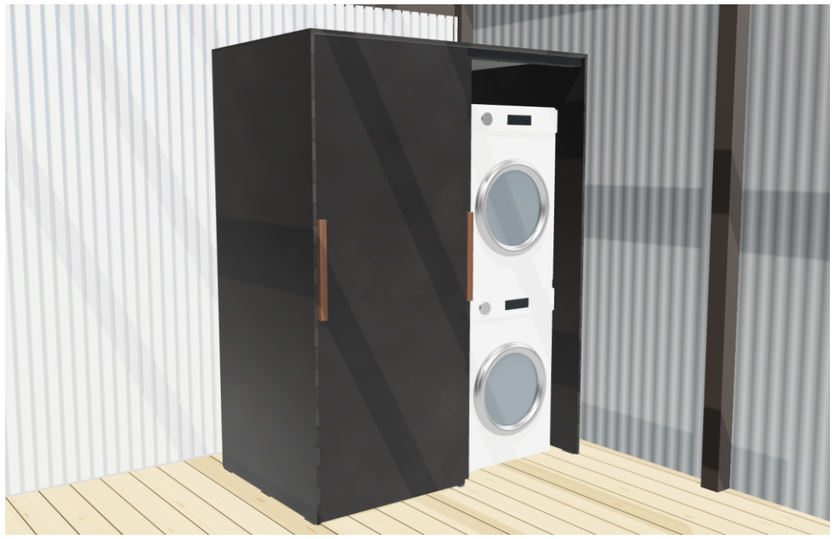
Outdoor appliance cabinet: fridge + stacked washer/dryer





A plain black formply box: flush top, a fridge door flush with the sides from 10 off the deck to 3 under the top, no front vents, and two short American walnut pulls.

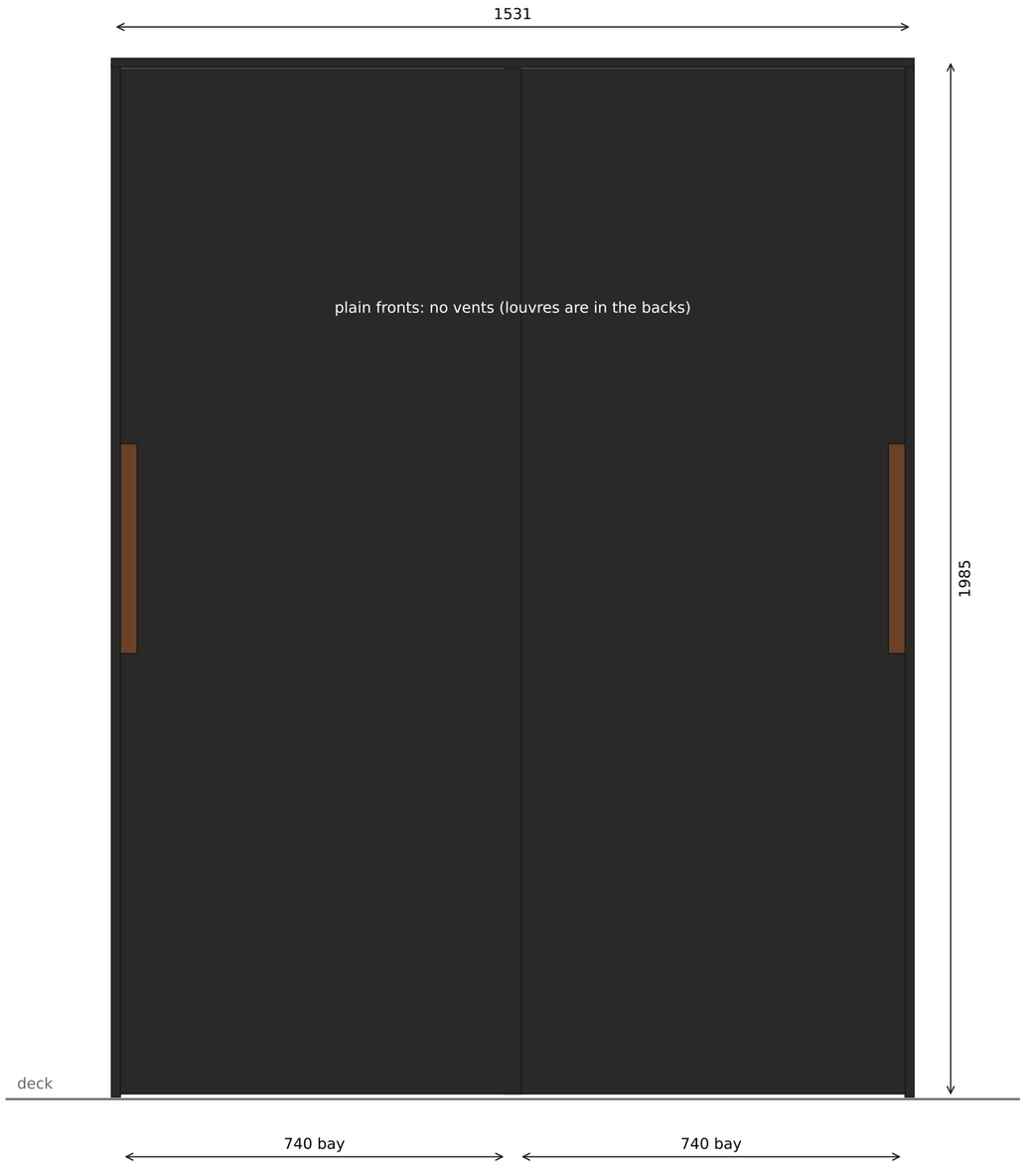
1531 W × 900 D × 1985 H. Two 740 clear bays. 8 CNC parts on 8 sheets of 2400 × 1200 × 17.

The fridge breathes through louvred openings low and high in the hidden back panels (about 230 cm2 net each level). Run the dryer with the laundry door slid open. The box sits on 4 mm ASA pads, held down by two stainless angles into the deck joists.

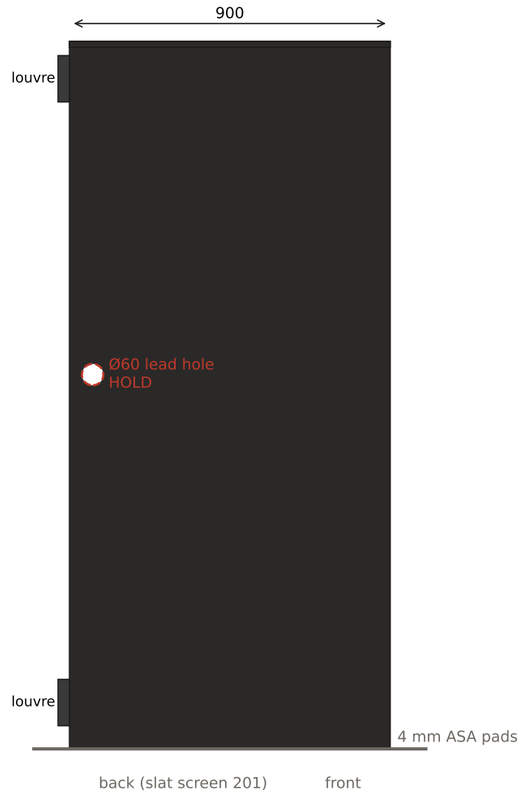
15 of 15 design checks pass (sheet 6). Open items must be closed before the cut.

General arrangement

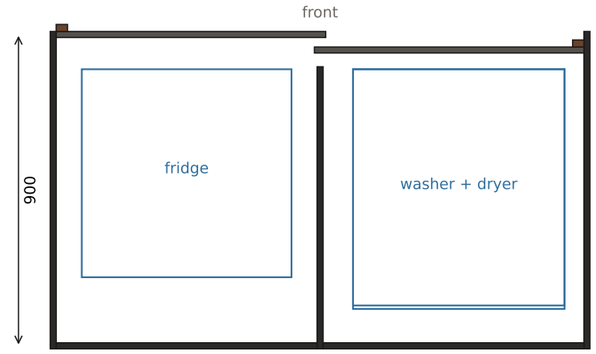
Front elevation (doors closed)



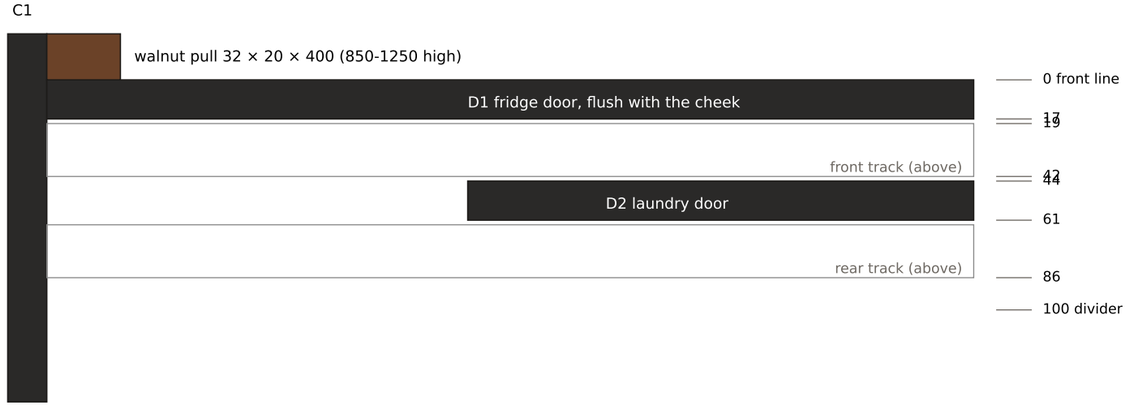
Side elevation (studio cheek, outside)



Plan (top removed)

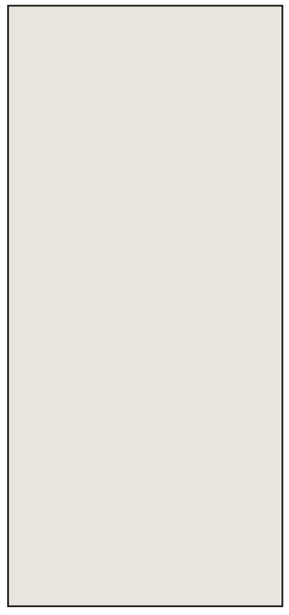


Section through the doors and tracks (plan, enlarged)



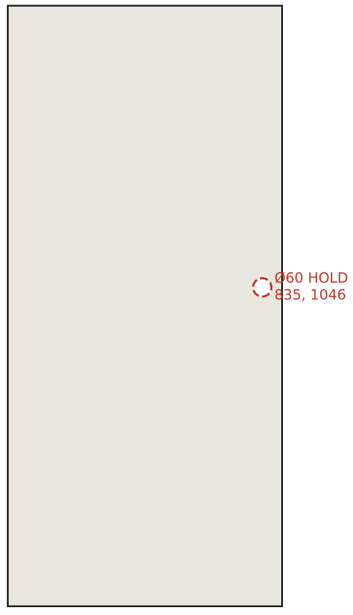
CNC part drawings

C1 Left cheek 900 × 1970 ×1



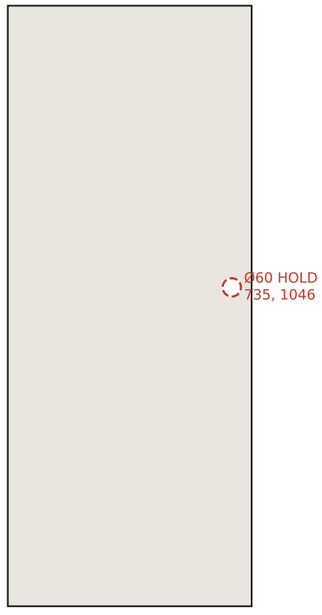
X = 0 at the cabinet front edge, Y = 0 at the bottom edge.

C3 Studio cheek 900 × 1970 ×1



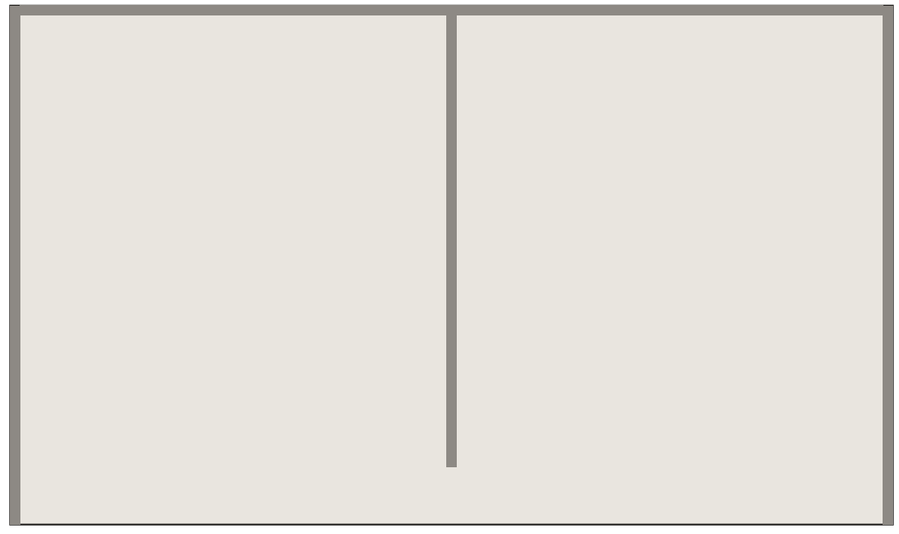
X = 0 at the cabinet front edge (mirror view: routed from the outside face). HOLD hole: confirm on site.

C2 Divider 800 × 1970 ×1



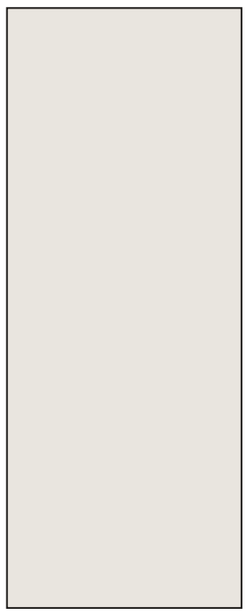
X = 0 at the front edge. HOLD hole: confirm on site.

C4 Top 1531 × 900 ×1



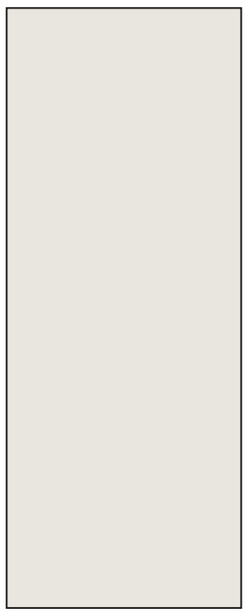
Routed on the UNDERSIDE: 6 mm housings for every vertical. X = 0 at the kitchen-side edge, Y = 0 at the front edge.

D1 Fridge door 764.5 × 1955 ×1



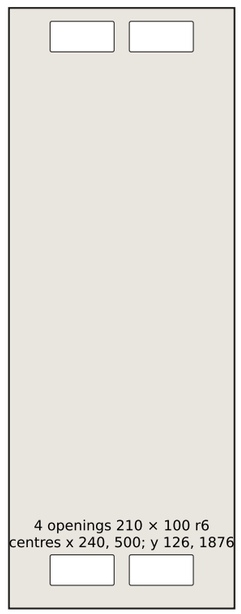
Plain panel. Bottom-edge groove 8 x 12 routed in the workshop, then sealed.

D2 Laundry door 764.5 × 1955 ×1



Plain panel. Bottom-edge groove 8 x 12 routed in the workshop, then sealed.

C5 Back, fridge bay 740 × 1970 ×1



4 openings 210 x 100, r6 corners, for the ASA louvres. Drawn from the inside face, X = 0 at the left.

C6 Back, laundry bay: as C5.
C4 is routed on its UNDERSIDE: 6 deep housings (grey) for C1, C2, C3, the backs and the head rail.
Cheek housings are 0.6 over the panel for glue and fit.

Cut list and indicative nests

Code	Part	Size	Qty	Notes
C1	Left cheek	900 × 1970	1	X = 0 at the cabinet front edge, Y = 0 at the bottom edge.
C2	Divider	800 × 1970	1	X = 0 at the front edge. HOLD hole: confirm on site.
C3	Studio cheek	900 × 1970	1	X = 0 at the cabinet front edge (mirror view: routed from the outside face). HOLD hole: confirm on site.
C4	Top	1531 × 900	1	Routed on the UNDERSIDE: 6 mm housings for every vertical. X = 0 at the kitchen-side edge, Y = 0 at the [...]
C5	Back, fridge bay	740 × 1970	1	4 openings 210 x 100, r6 corners, for the ASA louvres. Drawn from the inside face, X = 0 at the left.
C6	Back, laundry bay	740 × 1970	1	As C5.
D1	Fridge door	764.5 × 1955	1	Plain panel. Bottom-edge groove 8 x 12 routed in the workshop, then sealed.
D2	Laundry door	764.5 × 1955	1	Plain panel. Bottom-edge groove 8 x 12 routed in the workshop, then sealed.

S1 2400 × 1200 × 17



S2 2400 × 1200 × 17



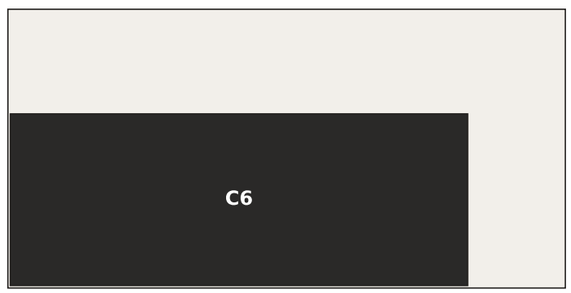
S3 2400 × 1200 × 17



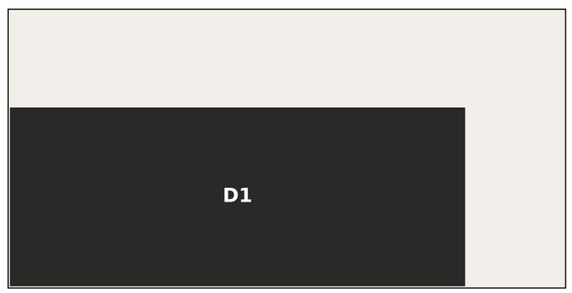
S4 2400 × 1200 × 17



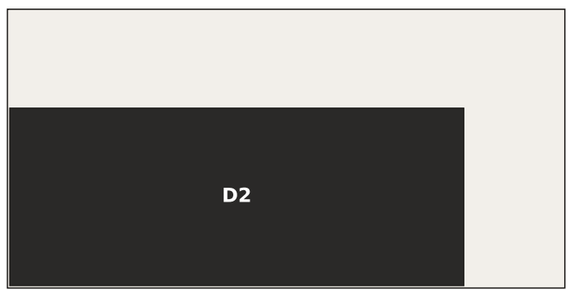
S5 2400 × 1200 × 17



S6 2400 × 1200 × 17



S7 2400 × 1200 × 17



S8 2400 × 1200 × 17



Build steps and shopping list

1. Confirm before cutting

Confirm the washer and dryer models and their stacking kit (dryer must be heat-pump or condenser). Choose the top-hung track kit (face-fixed hangers, two tracks under the top). Electrician confirms three outlets on the studio wall. Measure the deck and the gaps to the slat screen and the studio wall. Release the two HOLD lead holes, or leave them off the cut.

2. Seal, rout and print

Seal every CNC-cut edge with two coats of exterior edge sealer, bottom edges most of all. Rout the 8 x 12 guide groove along the bottom edge of each door and seal it. Print the ASA parts (Prints tab).

3. Pads on the deck

Mark the footprint. Bed the thirteen 4 mm ASA edge pads in sealant on the lines of the cheeks, divider and backs, one 10g x 25 each.

4. Carcass on the top, upside down

Lay the top film-face down on padded trestles, housings up. Run PU sealant in each housing. Stand C1, C2, C3, C5 and C6 in their housings; the housings square the box. Screw down through the top into each edge, 8g x 50 stainless at 200 centres, countersunk and sealed.

5. Angles and louvres

Fit the four black aluminium angles inside the back corners. Staple stainless mesh behind the eight back openings, then bed and screw an ASA louvre over each one from the outside, blades sloping down and out.

6. Stand it up and hold it down

Two people. Turn the carcass upright onto the pads. Check plumb and diagonals. Screw the two stainless hold-down angles inside the cheeks into the deck joists and the cheeks.

7. Tracks

Cut both top-hung tracks to 1497 and screw them up into the top: front track 19-42 back from the front edge, rear track 63-86.

8. Doors and pulls

Screw the walnut pulls on from behind: on the left edge of D1 and the right edge of D2, 850-1250 above the deck. Fit the face-fixed hangers and hang D1 on the front track and D2 on the rear. Set D1 flush with the cheek fronts, 10 off the deck and 3 under the top.

9. Floor guide

Close both doors. Screw the ASA guide to the deck under the crossover so its fins sit in the door grooves; run both doors end to end.

10. Grommets and leads

Only after the HOLD is released: fit the split grommets in the lead holes and route each original lead separately to the wall. No powerboard, nothing in a door track or under a machine foot.

11. Appliances in

Roll the fridge in, door hinge on the left, 107 back from the cabinet front. Stand the washer, fit the maker's stacking kit and lift the dryer on. Level all three on their own feet; the cabinet carries none of them.

Type	Item	Qty	Use
Sheet	Black film-faced formply 2400 x 1200 x 17, exterior (A-bond)	8 sheets	CNC supplier supplies and cuts (Cut list tab)
Sheet	Exterior edge sealer, black	1 L	Two coats on every cut edge and the door grooves
Sheet	PU sealant/adhesive, black (Sikaflex-11FC or similar)	4 x 300 ml	Housings, angles, louvres, pads
Hardware	Top-hung sliding track kit, black anodised, with face-fixed hangers, exterior rated, x 2 (one per door, about 20 kg each)	2 kits, cut to 1497	Must fit 23 deep x 40 high per track
Hardware	Aluminium angle 25 x 25 x 3, black anodised	4 x 1700	Inside the back corners
Hardware	Stainless angle 50 x 40 x 3	2 x 300	Hold-downs inside the cheeks
Hardware	Stainless insect mesh	1 m x 300	Behind the eight back openings
Timber	American walnut 32 x 20 x 400, from workshop offcuts	2	Door pulls
Timber	Exterior hardwood oil (e.g. Cutek or Intergrain UltraDeck Oil)	250 ml	Walnut pulls; seal end grain
Fixings	8g x 50 stainless countersunk timber screws	60	Through the top into the panel edges
Fixings	8g x 25 stainless countersunk	90	Corner and hold-down angles
Fixings	10g x 50 stainless countersunk	4	Hold-downs into the deck joists
Fixings	10g x 25 stainless countersunk	16	Pads and floor guide
Fixings	6g x 16 stainless pan	32	Louvres
Fixings	8g x 30 stainless countersunk	6	Pulls, from behind
Fixings	Stainless staples, 10 mm	1 box	Mesh
Print	ASA filament, black	about 1.5 kg	Prints tab: 4 STLs, 24 prints

Checks, open items and ASA prints

	Check	Result
Pass	Fridge side clearance (maker 50)	72.5 each side
Pass	Fridge back clearance (maker 50)	186 to the back panel: the chimney behind the fridge
Pass	Fridge top clearance (maker 100)	183 to the top; the tracks sit in front of the fridge face
Pass	Stack under the top	238 above the dryer top
Pass	Fridge passes the open door	opening 17-749.5, fridge 89.5-684.5
Pass	Laundry passes the open door	opening 841.5-1514, machines 859.0-1459.0
Pass	Rear pull clear of the closed front door	28 mm finger gap
Pass	Rear pull passes behind the front door	7 mm running gap
Pass	Rear door clear of the front track	2 mm
Pass	Rear track clear of the divider	14 mm
Pass	Front door hides both tracks	door tops 3 under the top
Pass	Fridge vents (Liebherr/Electrolux built-in rule: 200 cm2	420 cm2 gross per level per bay, about 231 net through louvre and mesh, pl
Pass	High vents above the appliances	high openings start 1830; fridge 1785, stack 1730
Pass	Doors clear of the deck	10 above the deck
Pass	Fridge door opens 115 deg past the cheek	closest 24 mm at 115 deg (simplified slab, left hinge)

Open item	Before the cut
Washer and dryer models	Provisional 600 x 680 x 850 and 600 x 670 x 850. Confirm both models and the maker's stacking kit; the dryer must be heat-pump or condenser.
Dryer air	Run the dryer with the laundry door slid open. Makers say a cupboard dryer needs louvred door openings or the door open while it runs (LG asks for two door openings of 387 cm2). The back louvres help but are not a substitute.
Power	Owner says there is a power point on the studio wall. Position, outlet count and weather cover are unmeasured. The two lead holes are HOLD until an electrician confirms.
Track kit	Needs face-fixed hangers so each door can run up in front of its own track. Each track must fit about 23 deep x 40 high under the top; the rear door clears the front track by only 2 mm on these numbers, so check the chosen kit's drawing.
Front track shows	Over the laundry half the front track sits in front of the set-back laundry door, so a 40 mm strip of it shows under the top. Specify black anodised track so it reads as a shadow line.
Rear door set-back	With bypass doors one door has to run behind the other: the laundry door sits 44 behind the flush fridge door. A fully flush front needs in-line flush-sliding hardware (both doors close in one plane); possible, but dearer and mostly rated for indoor use.
Door flatness	Without full-height stiles, 1955 tall formply doors rely on the balanced film faces to stay flat. Keep both faces sealed; if one bows, a black aluminium angle screwed up the back edge fixes it.
Site measure	Footprint, the 201 gap to the slat screen and the 195 gap to the studio wall come from the handover setout. Measure before the cut is released.

ASA print	Qty	Size	Notes
03_floor_guide.stl	1	28 × 70 × 19	Hidden floor guide under the door crossover. Two 6 mm fins ride 9 mm up into the 8 x 12 grooves in the door bottoms (doors are 10 above the deck), so nothing shows above the deck. 2 x 10g x 40 stainless into the deck.
04_cable_grommet.stl	2	82 × 82 × 20	Split grommet for the two 60 mm lead holes (C2, C3). Smooth bore protects the leads. Print flange down. HOLD: confirm the plugs fit through 48 mm or enlarge the hole first.
05_edge_pad.stl	13	80 × 24 × 4	4 mm pad under the ply edges (13 of them). The box reads as flush to the deck but the ply never sits in water and the deck boards can dry. Bed in sealant, 1 x 10g x 25 into the deck. Print 100% infill.
06_rear_louvre.stl	8	240 × 130 × 32	Louvre over each back opening (8). Blades slope down and out to shed rain. Staple stainless mesh behind the opening first, then bed the louvre in sealant and screw 4 x 6g x 16 through the flange. Print standing on its back flange; supports off.